EMBEDDED ML · COMPUTER VISION · FLIGHT CONTROL

A 92 mm drone that follows one enrolled person.

The circuit board is the airframe. Flight firmware on an ESP32-S3, perception off-board on a laptop over WiFi.

66 g **292 tests**

Not yet flown — boards not fabricated

PURSUIT-1 1

VERIFIED THIS WEEK

It can tell me from my friend.

400 frames, live webcam, two real people. My friend walks in, crosses in front of me, and stays after I leave.

98%

locked on me
0 frames locked on him

Never lost the lock. It never transferred.

This test had never been run before

METHOD

Split by identity, not by image.

I trained a face embedding on 4,324 real photographs. 127 people to train on, 31 held out and never seen.

93.9% if you split by image and let it see the same faces twice

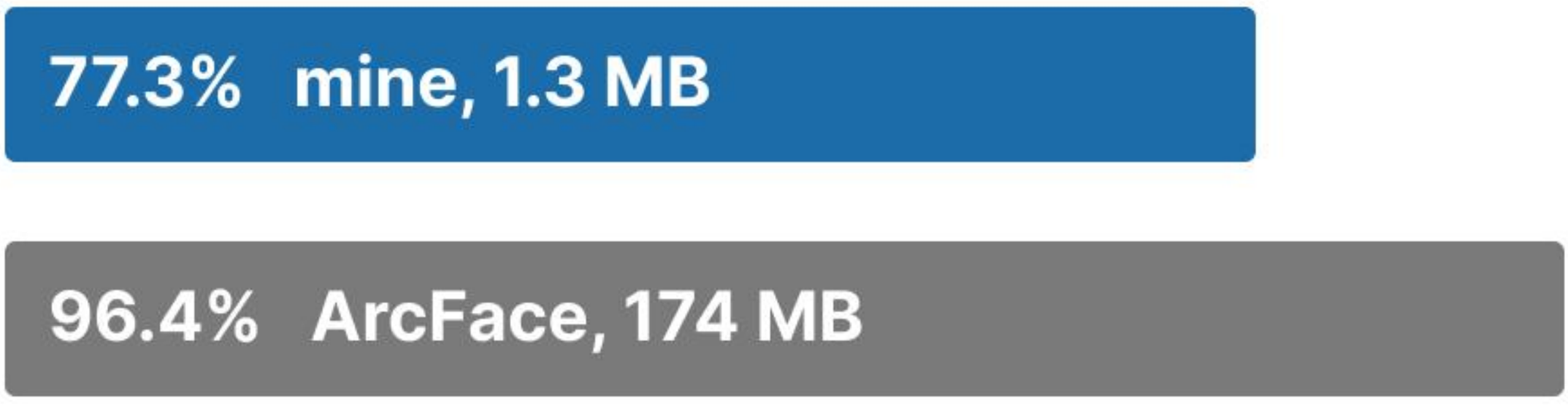
77.3% on people it has never seen the honest number

16.6 points of inflation, measured

BENCHMARK

136x the weights buys 19 points.

My 1.3 MB model against the 174 MB pretrained ArcFace, scored on identical image pairs.



Now I know what that dependency is worth.

ArcFace gets unaligned crops here — it scores 99.8% properly

CONTROL

The loop closes through the airframe.

Pitching to chase someone moves the camera, which moves where they appear in frame. That path has a gain of

34 / 26 = 1.31

Above 1, so the correction overshoots. Uncompensated it oscillates at the first frame of link delay — and the real round trip is 20 ms.

Found in simulation, before it could destroy the aircraft

WHAT IS NOT DONE

It has never flown.

The boards are not fabricated. The PID gains are scaled from typical values, not tuned on this airframe. The stability result is simulated, with estimated mass properties.

Everything above is bench or simulation, and every slide says which.

Code, write-up and every measurement

github.com/Pai-88/pursuit-drone

207 Python + 85 native C++ tests